MIL-F-1611E
17 September 1971

SUPERSEDING
MIL-F-1611D
16 March 1966

MILITARY SPECIFICATION

FAIRLEAD, TENT LINE, MAGNESIUM

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type and size of magnesium fairlead.

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification on the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-M-55	-	Magnesium Alloy, Permanent and Semi-Permanent Mold Castings.
QQ-M-56	-	Magnesium Alloy, Sand Castings.
TT-E-529	-	Enamel, Alkyd, Semi-gloss.
TT-P-666	-	Primer Coating, Zinc Yellow (Zinc Chromate), for Aluminum Surfaces.
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-M-3171	-	Magnesium Alloy, Processes for Pretreatment and Prevention of Corrosion on.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

FSC 8340

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.

DRAWING

ARMY NATICK LABORATORIES

5-4-234 - Fairlead, Tent Line, Magnesium.

(Miniature copy of Drawing 5-4-234, identified as figure 1, is attached for information purposes only.)

(Copies of specifications, standards, drawing, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 94 - Magnesium - Alloy Die Castings

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Associations, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 202, Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), the supplier shall furnish a sample for first article inspection and approval (see 4.2 and 6.3).

* 3.2 Materials.- Materials shall conform to the applicable referenced specifications, drawings, and requirements specified herein.

* 3.2.1 Magnesium alloy.- Magnesium alloy shall be produced by any one of the following methods:

(a) Sand casting. Sand casting shall conform to alloy AZ63A, AZ91C, or AZ92A, temper T-6 of QQ-M-56.

(b) Permanent mold casting. Permanent mold casting shall be in accordance with alloy AZ63A, AZ91C, or AZ92A, temper T-6 of QQ-M-55.

(c) Die casting. Die casting shall conform to magnesium alloy AZ91A or AZ91B of ASTM B 94.

3.2.2 Primer.- The primer shall conform to TT-P-666.

* 3.2.3 Enamel.- The enamel shall conform to class A or B, composition G or L, olive drab color No. 24087 of TT-E-529.

3.3 Construction.- Construction shall be in accordance with the requirements of Drawing 5-4-234 and as specified herein.

* 3.3.1 Casting.- The casting shall conform to any one of the three methods specified in 3.2.1 at the option of the manufacturer.

* 3.4 Finish.- Any burr, sharp edge and rough spot shall be removed prior to application of finish.

* 3.4.1 Chemical treatment.- All surfaces of the fairlead shall be given a chemical treatment conforming to type III or IV of MIL-M-3171.

* 3.4.2 Priming.- After the chemical treatment specified in 3.4.1, the fairlead shall be coated with primer specified in 3.2.2.

* 3.4.3 Enameling.- After the primer has been allowed to dry, the fairlead shall be given a coat of enamel specified in 3.2.3.

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:34Z
Check the source to verify that this is the current version before use.

3.5 Workmanship.- All surfaces shall be free from burrs, fins, rough spots, sharp edges, cracks, blow holes, voids or porous. The enamel coating shall level out to produce a smooth, uniform dry film without runs, wrinkles, drops, streaks, grit, coarse particles, and areas of thin or no film.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a first article is required, it shall be examined for defects in table I and specified dimensions. Any visual defect or dimension not within specified requirements shall be cause for rejection of the first article.

4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase document.

* 4.3.2 In-process inspection.- In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specified requirements. Whenever nonconformance is noted, correction shall be made to affected items and the process.

(a) Removal of burr, sharp edge and any rough spot is accomplished as specified in 3.4.

(b) Chemical treatment - Treatment of metal surfaces for painting is accomplished as specified in 3.4.1.

(c) Priming of metal - Metal surfaces have been coated with primer as specified in 3.4.2.

(d) Drying of primer - Primer has been allowed to dry prior to enameling as specified in 3.4.3.

Source: https://assist.dla.mil -- Downloaded: 202640-16T13:34Z
Check the source to verify that this is the currentversion before use.

4.3.3 End item inspection.- The lot shall be all of the fairleads offered for inspection at one time. The sample unit for this examination shall be one finished fairlead.

* 4.3.3.1 Visual examination.- Examination shall be made of the fairleads for the defects in table I. The inspection level shall be II with an acceptable quality level (AQL) of 4.0 for major defects and 10.0 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Classification of defects

Examine	Defect	Classification Major	Minor
Finish	Not finished	X	
	Color not as specified.		X
	Coating not smooth or uniform		X
	Finish is not dry, i.e., wet or tacky to touch		X
	Runs, wrinkles, drops, streaks, grit, or coarse particles		X
	Area of no film exposing bare metal	X	
	Area of thin film exposing primer coating		X
Construction and workmanship	Not fabricated as specified, such as hole missing	X	
	Burr, fin, rough spot 1/		X
	Sharp edges, cracks, blow holes, voids, or porous 1/		X

1/ Major - if in area which could abrade tent line or result in injury to user.

4.3.3.2 Dimensional examination. - Inspection shall be made for defects in dimensions. Any dimension not within the specified requirements shall be classified as a defect. The inspection level shall be S-2 with an AQL of 4.0, expressed in terms of defects per hundred units.

4.3.4 Examination of preparation for delivery.- An examination shall be made to determine that packaging, packing and markings are in compliance with section 5 of this specification. Defects shall be as specified in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

Source: https://assist.dla.mil -- Downloaded: 202640-16T13:34Z
Check the source to verify that this is the current version before use.

TABLE II.- Examination of preparation for delivery

Examine	Defect
Markings (exterior and interior container)	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.
Material	Any nonconforming component, component missing, damaged, or otherwise defective.
Workmanship	Bulged or distorted container.
Contents	Quantity of fairleads per package or shipping container, as applicable, is more or less than required.

* 4.3.5 Examination of shipping containers.- When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, strapping or tape banding shall be made in accordance with the appendix of PPP-B-636.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

* 5.1.1 Level A.- One hundred fairleads shall be packaged in a close-fitting intermediate fiberboard box conforming to type CF, variety SW, or type SF, class domestic, grade 200 of PPP-B-636. Each box shall have the contents agitated during filling to assure a compact package and minimum void space. Closure shall be with gummed Kraft or pressure sensitive tape and in accordance with the appendix of the box specification.

* 5.1.2 Level C.- Fairleads shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

* 5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

* 5.2.1 Level A.- Four hundred fairleads, packaged as specified in 5.1, shall be packed in a close-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of PPP-B-636.

Source: https://assist.dla.mil -- Downloaded: 202610-16T13:34Z
Check the source to verify that this is the current version before use.

* 5.2.2 Level B.- Four hundred fairleads, packaged as specified in 5.1, shall be packed in a close-fitting fiberboard shipping container conforming to style RSC, type CF, variety SW, or type SF, class domestic, grade 275 of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636.

* 5.2.2.1 When specified (see 6.2), the shipping container shall be grade V3c, V3s or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

* 5.2.3 Level C.- Fairleads, packaged as specified, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The fairlead covered by this specification is intended to provide a means of attaching angular guide lines to a tent eave.

* 6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.2, and 6.3).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(d) Whether weather-resistant fiberboard shipping containers are required for level B shipment (see 5.2.2.1).

* 6.3 First article.- When a first article is required, it shall be inspected and approved under the appropriate provisions of ASPR 7-104.55. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.

Custodians:

Army - GL
Navy - SA
Air Force - 82

User activities:

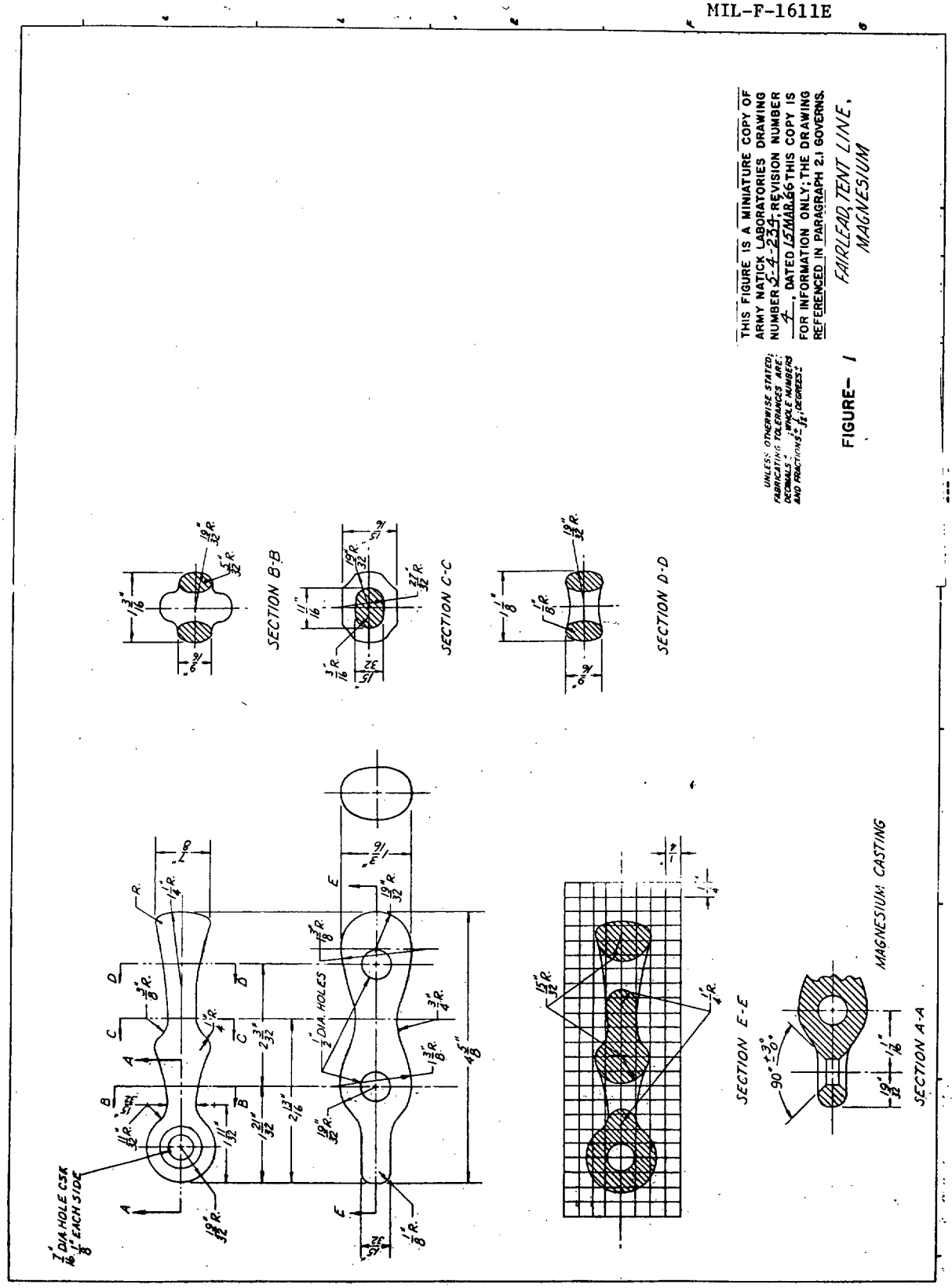
Army - ME
Navy - YD

Preparing activity:

Army - GL

Project No. 8340-0277

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.



☆ U. S. GOVERNMENT PRINTING OFFICE: 1971-714-158/2491

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

FAIRLEAD, TENT LINE, MAGNESIUM MIL-F-1611E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

Source: https://assist.dla.mil -- Downloaded: 2026-10-16T13:34Z
Check the source to verify that this is the current version before use.